MARK SCHEME · TEACHER COPY

11BR · Topic 8: Databases

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Total marks: 73

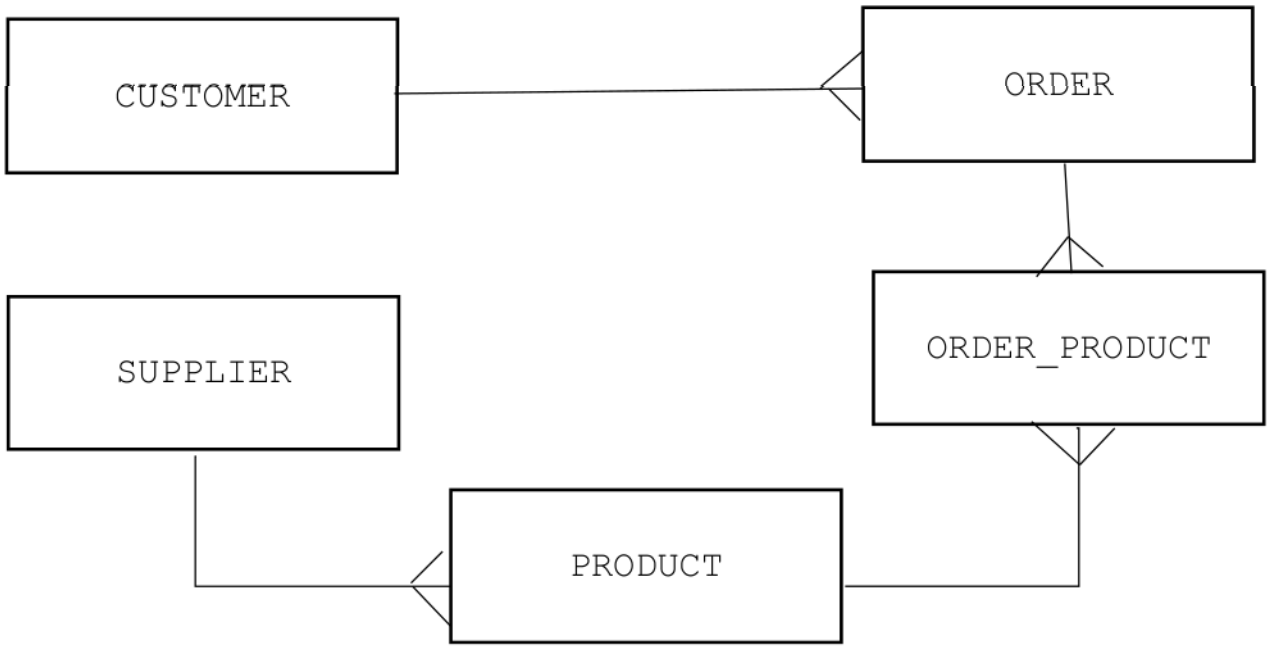
Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 73 marks

Question 1 · May/June 2025 Paper 13, original Q6 (a),(b),(c),(d) [18 marks]

Question	Answer	Marks
1(a)	**1 mark** for each correct relationship 	**4**

1(b)	**1 mark** each • INSERT INTO PRODUCT • VALUES with opening and closing brackets • Inserting ProductID, ProductName and SupplierID correctly as strings including quotation marks into correct fields • Inserting quantity and cost correctly into correct fields Example 1: INSERT INTO PRODUCT VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223"); Example 2: INSERT INTO PRODUCT (ProductID, ProductName, QuantityInBox, Cost, SupplierID) VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223");	**4**

1(c)	**1 mark** each • Selection of customer name and counting any field from ORDER as an appropriate identifier • Joining tables ORDER and CUSTOMER • AND (or WHERE) clause: Collected = FALSE • Grouping by customer ID or customer name Example 1: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER, CUSTOMER WHERE ORDER.CustomerID = CUSTOMER.CustomerID AND Collected = FALSE GROUP BY CUSTOMER.CustomerID Example 2: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER INNER JOIN CUSTOMER ON ORDER.CustomerID = CUSTOMER.CustomerID WHERE Collected = FALSE GROUP BY CUSTOMER.CustomerID	**4**
1(d)(i)	**1 mark** each to **max 3** e.g. • Relationships • Views • Data types • Validation rules	**3**

1(d)(ii)	**1 mark** each to **max 3** e.g. • To create / modify / delete database objects • To create a form for data input • To add tools to a form • for example, drop-down boxes / buttons etc. • To design a report to show the output in an organised manner • To add a menu to enable users to choose different actions / run different queries	**3**

Question 2 · May/June 2026 Paper 12, original Q3 (a),(b),(c) [16 marks]

Question	Answer	Marks	Guidance
2	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		
2(a)	**1 mark** for each correct relationship • The relationship between `PLAYER` and `CHARACTER` is one to many /1:M // one player can have many characters, **but** each character can only be created by one player • The relationship between `GROUP` and `CHARACTER` is many to many / M:M // one group can have many characters, **and** one character can be in many groups	**2**	The second option in each case needs both 'halves' Accept a m:m relationship between `PLAYER` and `GROUP` Accept user for player Mark the writing first, but accept accurate E-R diagram(s), if needed for clarification

2(b)(i)	**1 mark** for each correct line number **and** the appropriate correction • Line 02 • `CharacterID` **`VARCHAR`**, // `CharacterID` **`VARCHAR(5)`**, • Line 05 • **`Level`** `INT`, • Line 08 • `PRIMARY KEY(CharacterID)` • Line 09 • `FOREIGN KEY(PlayerID)` **`REFERENCES PLAYER(PlayerID)`**	**4**	Ignore extra `NOT NULL` on line 02 but not on line 05 or line 08

2(b)(ii)	**1 mark** for each correct point
	
	• Selecting the player's name and counting a field
	• Using an appropriate name
	• Linking or joining the correct tables
	• Grouping the results
	• Ordering in descending order by player ID
	
	Example 1:
	`SELECT PlayerName, COUNT(CHARACTER.CharacterID)` MP1
	`AS TotalCharacters` MP2
	`FROM CHARACTER, PLAYER`
	`WHERE CHARACTER.PlayerID = PLAYER.PlayerID` MP3
	`GROUP BY PlayerName` MP4
	`ORDER BY PLAYER.PlayerID DESC;` MP5
	
	Example 2:
	`SELECT PlayerName, COUNT(CharacterID)` MP1
	`AS TotalCharacters` MP2
	`FROM CHARACTER INNER JOIN PLAYER`
	`ON CHARACTER.PlayerID = PLAYER.PlayerID` MP3
	`GROUP BY PlayerName` MP4
	`ORDER BY PLAYER.PlayerID DESC;` MP5
		5	Accept any correct method of joining the tables, but no mix and match between methods
			
			The only attribute that needs a table reference is the PlayerID. Ignore other table references as long as the correct table is used
			
			Take care with the DESC

2(c)	**1 mark** for **two** data dictionary examples For example: • data types **of individual fields/attributes** • type of validation • primary key • foreign key **1 mark** for each remaining term (**4 marks**) • data dictionary • logical • access rights • developer interface The **data dictionary** stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples that will be stored are **example 1** and **example 2** When designing a database, the **logical** schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS). A DBMS provides security features to protect the data. **Access rights** are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data. When creating a database a **developer interface** provides tools to create visual components for example forms and reports.	**5**	Mark the first answer in each space Do not accept table names or field names or relationships, they are given in the question There are many answers for the examples, the list is suggestions only Do not accept metadata as an example or instead of data dictionary Just 'key' is NE for primary or foreign keys Just data type is TV for the second example

Question 3 · Oct/Nov 2023 Paper 12, original Q2 (a),(b),(c) [11 marks]

Question	Answer	Marks

3(a)	**1 mark** for each term (**max 3**) Entity: • An object about which data can be stored Primary key: • The **unique** attribute / combination of attributes used to identify the **record** / **tuple** Referential integrity: • Makes sure that if data is changed in one place the change is reflected in all related records - cascading update/delete • Makes sure that data that does not exist cannot be referenced • Ensures that every foreign key has a **corresponding** primary key // A logical dependency of a foreign key on a primary key • Ensures that the data in the database is consistent / up to date • Prevents records from being added/deleted/modified incorrectly • Makes sure any queries return accurate and complete results	**3**

3(b)	**1 mark** for each bullet point (**max 4**) **Max 2** if no descriptions • Backup / recovery procedures • … automatically takes copies of the database and store off site on a regular basis / weekly, etc. • ... so that the data can be recovered if lost • Use of access rights • … some users are given different access permissions to different tables • ... read/write, read only, full access, etc. • Views • … different users are able to see different parts of the database • ... only see what users need to see // by example • Record and table locking • … prevents simultaneous access to data • ... so updates are not lost // data is not overwritten • Encryption • ... the data is turned into ciphertext • ... so it cannot be understood **without a decryption key**	**4**

3(c)	**1 mark** for each bullet point (**max 4**) • Identify **repeating** groups of attributes ... • ... Subject **and** SubjectCode • Ensure each field is atomic • ... StudentName should be split into e.g. FirstName and LastName • Identify the primary key for the table	**4**

Question 4 · Oct/Nov 2024 Paper 12, original Q6 (a),(b) [12 marks]

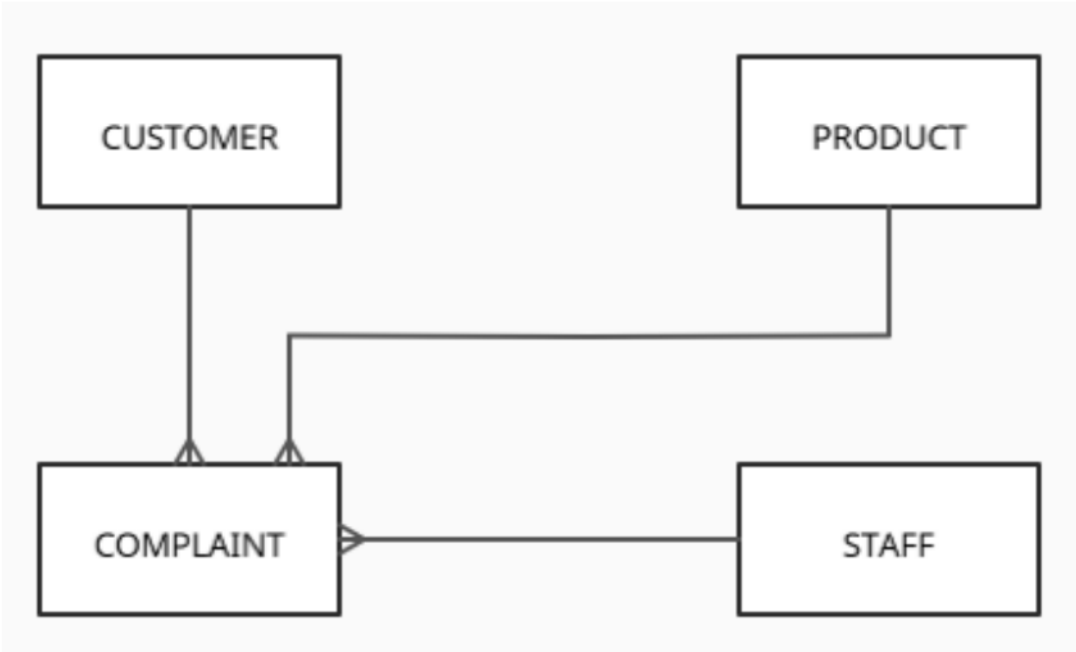
Question	Answer	Marks
4(a)	**1 mark** each to **max 3** e.g. • There is reduced data redundancy // less repeated data • … because each item of data is only stored once • Data consistency is maintained // Data integrity is improved • … changes in one table will automatically update in another • ... linked data cannot be entered differently in two tables • Program-data independence is ensured • … changes to the data do not require programs to be re-written // queries are not dependent on the structure of the data • Complex queries are easier to run • Different views can be provided • …. so users can only see specific aspects of the database • Multiple concurrent access is possible • ... through record locking	**3**

4(b)(i)	**1 mark** each to **max 6** • CUSTOMER to JOB is 1 to many • … implemented by Primary Key in CUSTOMER is Foreign Key in JOB • EMPLOYEE to LOGIN_DATA is 1 to 1 • … implemented by Primary Key in EMPLOYEE is Foreign Key in LOGIN_DATA • JOB to JOB_EMPLOYEE is 1 to many • … implemented by Primary Key in JOB is Foreign Key in JOB_EMPLOYEE • EMPLOYEE to JOB_EMPLOYEE is 1 to many • … implemented by Primary Key in EMPLOYEE is Foreign Key in JOB_EMPLOYEE	**6**

4(b)(ii)	**1 mark** each
	
	• Select SUM of Amount
	• From the correct table **and** one correct condition
	• Remaining correct condition
	
	Example:
	
	`SELECT SUM(Amount)`
	`FROM INVOICE`
	`WHERE Paid = "Y"`
	`AND DateSent >= #01/01/2023# AND DateSent <=`
	`#31/12/2023#`
		3

Question 5 · Oct/Nov 2025 Paper 13, original Q5 (a),(b),(c),(d) [16 marks]

Question	Answer	Marks

5(a)	**1 mark** for each correct relationship, **max 3** marks 	**3**

5(b)	**1 mark** per bullet point, **max 4** marks
	
	• Create table with opening and closing brackets and commas separating the attributes
	• Appropriate data types for StaffFirstName, StaffLastName and Department
	• Appropriate data types for StaffID and RemoteWorker
	• Primary key correctly defined
	
	Example Answer 1:
	`CREATE TABLE STAFF(`
	`StaffID INTEGER,`
	`StaffFirstName VARCHAR,`
	`StaffLastName VARCHAR,`
	`Department CHAR,`
	`RemoteWorker BOOLEAN,`
	`PRIMARY KEY (StaffID)`
	`);`
	
	Example Answer 2:
	`CREATE TABLE STAFF(`
	`StaffID INTEGER NOT NULL PRIMARY KEY,`
	`StaffFirstName VARCHAR,`
	`StaffLastName VARCHAR,`
	`Department CHAR,`
	`RemoteWorker BOOLEAN`
	`);`
		4

5(c)	**1 mark** per bullet point, **max 5** marks
	
	• SELECT and the correct attributes
	• FROM and the correct tables
	• Tables joined correctly
	• Correct condition for the rating
	• Correct ORDER BY clause
	
	Example Answer 1
	`SELECT PRODUCT.ProductID, ProductName, ComplaintDetails`
	`FROM PRODUCT, COMPLAINT`
	`WHERE PRODUCT.ProductID = COMPLAINT.ProductID`
	`AND Rating <=5`
	`ORDER BY Rating DESC;`
	
	Example Answer 2
	`SELECT PRODUCT.ProductID, ProductName, ComplaintDetails`
	`FROM PRODUCT INNER JOIN COMPLAINT`
	`ON PRODUCT.ProductID = COMPLAINT.ProductID`
	`WHERE Rating <=5`
	`ORDER BY Rating DESC;`
		5

5(d)	**One mark** for identification of method **one mark** for corresponding description **max 4** marks • Authentication methods / passwords / biometrics / 2-factor authentication can be implemented • ... which prevents unauthorised access to the customer's data • Access rights / privileges can be set • ... so that only those with correct permissions can read / edit customer's data • Regular backups can be scheduled • ... so that a second copy of the customer's data is available in case of loss/damage • The data can be encrypted • ... so that the customer's data cannot be understood by anyone who gains unauthorized access • Different views can be created • ... so that not everyone can see the customer's data	**4**